SDG 2030 | India's Development Trajectories

A time-series analysis of maternal mortality, sanitation and clean cooking

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Overview

This project examines India's historical progress across three development indicators and uses time-series models to project their trajectories towards 2030.

Indicator	2030 benchmark
Maternal mortality ratio (MMR)	70 deaths per 100,000 live births
Safely managed sanitation	100%
Clean cooking access	100%

Forecasts are conditional projections based on historical patterns, not guarantees of future outcomes.

1. Data

```
data_file <- "India_SDG_TimeSeries_RawData.xlsx"

if (!file.exists(data_file)) {
  stop(
    "Data workbook not found. Place India_SDG_TimeSeries_RawData.xlsx ",
    "in the same folder as this R Markdown file."
  )
}

df <- read_excel(data_file, sheet = "Raw_Data") %>%
  mutate(
    Year = as.integer(Year),
    MMR = as.numeric(MMR),
    `Safely Managed Sanitation (%)` =
      as.numeric(`Safely Managed Sanitation (%)`),
    `Clean Cooking (%)` =
      as.numeric(`Clean Cooking (%)`)
  ) %>%
  arrange(Year)

required_cols <- c(
  "Year", "MMR",
  "Safely Managed Sanitation (%)",
```

```
  "Clean Cooking (%)"
)

if (!all(required_cols %in% names(df))) {
  stop("The workbook does not contain all expected columns.")
}

if (anyDuplicated(df$Year)) {
  stop("Duplicate years found. Check the source data.")
}

if (!all(diff(df$Year) == 1)) {
  stop("Years must be consecutive. Check the source data.")
}

if (anyNA(df$Year) ||
    anyNA(df$`Safely Managed Sanitation (%)`) ||
    anyNA(df$`Clean Cooking (%)`)) {
  stop("Missing years or percentage values detected.")
}

if (sum(!is.na(df$MMR)) < 10) {
  stop("Insufficient observed MMR values.")
}

kable(
  tibble(
    Measure = c(
      "First year", "Last year",
      "Observed MMR values", "Missing MMR values"
    ),
    Value = c(
      min(df$Year),
      max(df$Year),
      sum(!is.na(df$MMR)),
      sum(is.na(df$MMR))
    )
  ),
  caption = "Dataset summary"
)
```

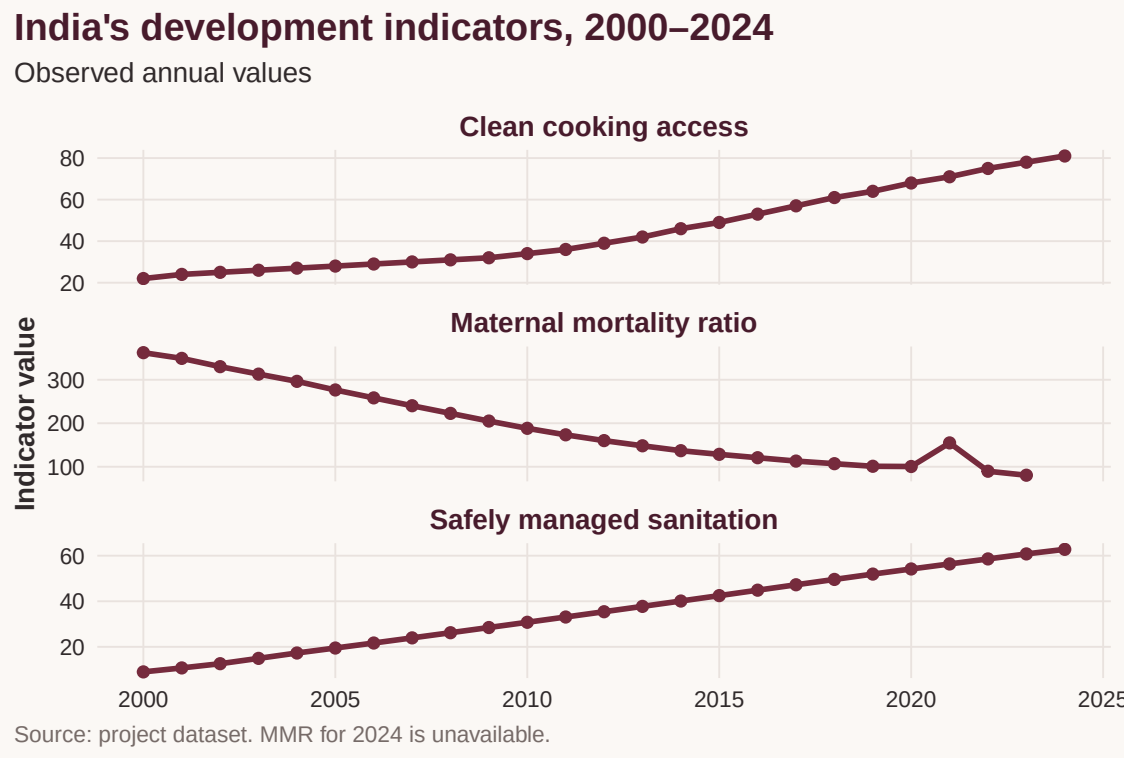
Table 1: Dataset summary

Measure	Value
First year	2000
Last year	2024
Observed MMR values	24
Missing MMR values	1

2. Historical trajectories

```
historical <- df %>%
  pivot_longer(
    cols = c(
      MMR,
      `Safely Managed Sanitation (%)`,
      `Clean Cooking (%)`
    ),
    names_to = "Indicator",
    values_to = "Value"
  ) %>%
  mutate(
    Indicator = recode(
      Indicator,
      "MMR" = "Maternal mortality ratio",
      "Safely Managed Sanitation (%)" =
        "Safely managed sanitation",
      "Clean Cooking (%)" = "Clean cooking access"
    )
  )

ggplot(historical, aes(Year, Value)) +
  geom_line(
    colour = sdg_burgundy,
    linewidth = 1.05,
    na.rm = TRUE
  ) +
  geom_point(
    colour = sdg_burgundy,
    size = 1.8,
    na.rm = TRUE
  ) +
  facet_wrap(
    ~ Indicator,
    scales = "free_y",
    ncol = 1
  ) +
  scale_x_continuous(
    breaks = pretty_breaks(n = 7)
  ) +
  labs(
    title = "India's development indicators, 2000-2024",
    subtitle = "Observed annual values",
    x = NULL,
    y = "Indicator value",
    caption = "Source: project dataset. MMR for 2024 is unavailable."
  ) +
  theme_sdg()
```



The indicators use different units and are displayed on separate scales. MMR is a mortality rate; sanitation and clean cooking are percentages.

3. Forecast model validation

Models are trained on 2000–2018 and evaluated against observed values from 2019–2023. Lower MAE and RMSE indicate better test-period accuracy.

```
train_data <- df %>%
  filter(Year >= 2000, Year <= 2018) %>%
  arrange(Year)

test_data <- df %>%
  filter(Year >= 2019, Year <= 2023) %>%
  arrange(Year)

if (nrow(train_data) != 19 || nrow(test_data) != 5) {
  stop("Unexpected training or test period. Check the dataset.")
}

evaluate_models <- function(variable, indicator) {

  y_train <- train_data[[variable]]
  y_test <- test_data[[variable]]

  if (anyNA(y_train) || anyNA(y_test)) {
    stop(paste("Missing values in validation series:", indicator))
```

```
  }

  h <- length(y_test)

  naive_pred <- naive(
    ts(y_train, frequency = 1), h = h
  )$mean

  drift_pred <- rwf(
    ts(y_train, frequency = 1),
    h = h,
    drift = TRUE
  )$mean

  linear_fit <- lm(
    y ~ t,
    data = data.frame(
      y = y_train,
      t = seq_along(y_train)
    )
  )

  linear_pred <- predict(
    linear_fit,
    newdata = data.frame(
      t = length(y_train) + seq_len(h)
    )
  )

  ets_pred <- forecast(
    ets(ts(y_train, frequency = 1)),
    h = h
  )$mean

  predictions <- list(
    Naive = as.numeric(naive_pred),
    Drift = as.numeric(drift_pred),
    `Linear trend` = as.numeric(linear_pred),
    ETS = as.numeric(ets_pred)
  )

  bind_rows(lapply(names(predictions), function(model_name) {
    errors <- y_test - predictions[[model_name]]

    tibble(
      Indicator = indicator,
      Model = model_name,
      MAE = mean(abs(errors)),
      RMSE = sqrt(mean(errors^2))
    )
  }))
}
```

```
accuracy_results <- bind_rows(
  evaluate_models("MMR", "Maternal mortality ratio"),
  evaluate_models(
    "Safely Managed Sanitation (%)",
    "Safely managed sanitation"
  ),
  evaluate_models("Clean Cooking (%)", "Clean cooking access")
)

kable(
  accuracy_results %>%
    arrange(Indicator, MAE) %>%
    mutate(
      MAE = round(MAE, 2),
      RMSE = round(RMSE, 2)
    ),
  caption = "Out-of-sample forecast accuracy, 2019-2023"
)
```

Table 2: Out-of-sample forecast accuracy, 2019–2023

Indicator	Model	MAE	RMSE
Clean cooking access	ETS	1.93	2.09
Clean cooking access	Drift	3.70	4.16
Clean cooking access	Naive	10.20	11.34
Clean cooking access	Linear trend	10.73	10.93
Maternal mortality ratio	ETS	14.89	29.01
Maternal mortality ratio	Naive	20.82	25.95
Maternal mortality ratio	Drift	40.84	49.45
Maternal mortality ratio	Linear trend	67.58	73.31
Safely managed sanitation	Drift	0.06	0.06
Safely managed sanitation	ETS	0.27	0.33
Safely managed sanitation	Linear trend	0.29	0.31
Safely managed sanitation	Naive	6.78	7.47

The validation results support ETS for MMR and clean cooking, and the Drift model for sanitation. The final forecasts below use these model choices.

4. Forecasts to 2030

MMR is fitted using observations through 2023 because its 2024 value is missing. Sanitation and clean cooking are fitted through 2024.

```
forecast_end <- 2030L

# MMR: last observed year is 2023
mmr_data <- df %>%
  filter(!is.na(MMR)) %>%
  arrange(Year)
```

```r
mmr_last_year <- max(mmr_data$Year)

mmr_model <- ets(
  ts(
    mmr_data$MMR,
    start = min(mmr_data$Year),
    frequency = 1
  )
)

mmr_fc <- forecast(
  mmr_model,
  h = forecast_end - mmr_last_year,
  level = 95
)

# Sanitation: Drift model
sanitation_data <- df$`Safely Managed Sanitation (%)`
sanitation_last_year <- max(df$Year)

sanitation_model <- rwf(
  ts(
    sanitation_data,
    start = min(df$Year),
    frequency = 1
  ),
  h = forecast_end - sanitation_last_year,
  drift = TRUE,
  level = 95
)

# Clean cooking: ETS
cooking_data <- df$`Clean Cooking (%)`
cooking_last_year <- max(df$Year)

cooking_model <- ets(
  ts(
    cooking_data,
    start = min(df$Year),
    frequency = 1
  )
)

cooking_fc <- forecast(
  cooking_model,
  h = forecast_end - cooking_last_year,
  level = 95
)

# Assemble forecast values with explicit calendar years
forecast_data <- bind_rows(
  tibble(
    Year = seq(mmr_last_year + 1, forecast_end),
```

```
    Indicator = "Maternal mortality ratio",
    Forecast = as.numeric(mmr_fc$mean),
    Lower = as.numeric(mmr_fc$lower[, "95%"]),
    Upper = as.numeric(mmr_fc$upper[, "95%"])
  ),
  tibble(
    Year = seq(sanitation_last_year + 1, forecast_end),
    Indicator = "Safely managed sanitation",
    Forecast = as.numeric(sanitation_model$mean),
    Lower = as.numeric(sanitation_model$lower[, "95%"]),
    Upper = as.numeric(sanitation_model$upper[, "95%"])
  ),
  tibble(
    Year = seq(cooking_last_year + 1, forecast_end),
    Indicator = "Clean cooking access",
    Forecast = as.numeric(cooking_fc$mean),
    Lower = as.numeric(cooking_fc$lower[, "95%"]),
    Upper = as.numeric(cooking_fc$upper[, "95%"])
  )
)

targets <- tibble(
  Indicator = c(
    "Maternal mortality ratio",
    "Safely managed sanitation",
    "Clean cooking access"
  ),
  Target = c(70, 100, 100)
)

forecast_2030 <- forecast_data %>%
  filter(Year == 2030) %>%
  left_join(targets, by = "Indicator") %>%
  mutate(
    Forecast = round(Forecast, 1),
    Lower = round(Lower, 1),
    Upper = round(Upper, 1),
    Target = round(Target, 1),
    Assessment = case_when(
      Indicator == "Maternal mortality ratio" &
        Forecast <= Target ~ "Target met in point forecast",
      Indicator == "Maternal mortality ratio" &
        Forecast > Target ~ "Target not met in point forecast",
      Indicator != "Maternal mortality ratio" &
        Forecast >= Target ~ "Target met in point forecast",
      TRUE ~ "Target not met in point forecast"
    )
  )

kable(
  forecast_2030 %>%
    select(
      Indicator, Forecast, Lower, Upper,
```

```
      Target, Assessment
    ),
  col.names = c(
    "Indicator", "2030 forecast",
    "Lower 95%", "Upper 95%",
    "Target", "Point-forecast assessment"
  ),
  caption = "Projected 2030 values and 95% prediction intervals"
)
```

Table 3: Projected 2030 values and 95% prediction intervals

Indicator	2030 forecast	Lower 95%	Upper 95%	Target	Point-forecast assessment
Maternal mortality ratio	94.3	7.0	181.7	70	Target not met in point forecast
Safely managed sanitation	76.2	75.4	77.1	100	Target not met in point forecast
Clean cooking access	99.0	90.9	107.2	100	Target not met in point forecast

Combined forecast figure

The three panels share the same visual design but use independent y-axes because the indicators have different units. The burgundy line shows observations, the maroon line shows forecasts, the shaded region shows the 95% prediction interval, and the brown dashed line marks the target.

```
forecast_plot <- forecast_data %>%
  mutate(Series = "Forecast")

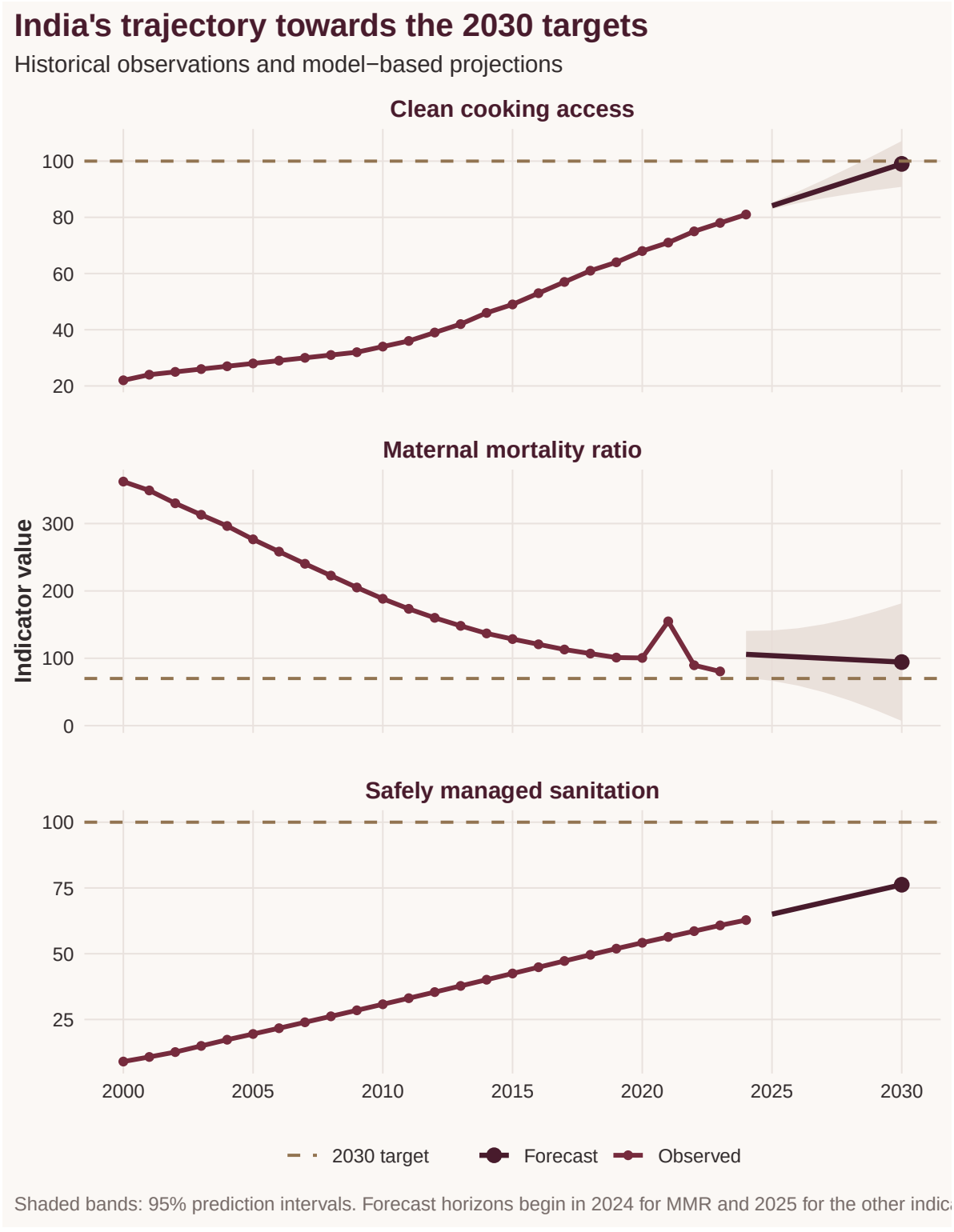
observed_plot <- historical %>%
  mutate(Series = "Observed")

target_plot <- targets

ggplot() +
  geom_ribbon(
    data = forecast_plot,
    aes(
      x = Year,
      ymin = Lower,
      ymax = Upper
    ),
    fill = sdg_neutral,
    alpha = 0.85
  ) +
  geom_line(
    data = observed_plot,
    aes(
      x = Year,
      y = Value,
```

```
      colour = "Observed"
    ),
    linewidth = 1.05,
    na.rm = TRUE
  ) +
  geom_point(
    data = observed_plot,
    aes(
      x = Year,
      y = Value,
      colour = "Observed"
    ),
    size = 1.5,
    na.rm = TRUE
  ) +
  geom_line(
    data = forecast_plot,
    aes(
      x = Year,
      y = Forecast,
      colour = "Forecast"
    ),
    linewidth = 1.15
  ) +
  geom_point(
    data = forecast_plot %>% filter(Year == 2030),
    aes(
      x = Year,
      y = Forecast,
      colour = "Forecast"
    ),
    size = 2.8
  ) +
  geom_hline(
    data = target_plot,
    aes(
      yintercept = Target,
      linetype = "2030 target"
    ),
    colour = sdg_brown,
    linewidth = 0.7
  ) +
  facet_wrap(
    ~ Indicator,
    scales = "free_y",
    ncol = 1
  ) +
  scale_colour_manual(
    values = c(
      "Observed" = sdg_burgundy,
      "Forecast" = sdg_maroon
    )
  ) +
```

```r
  scale_linetype_manual(
    values = c("2030 target" = "dashed")
  ) +
  scale_x_continuous(
    breaks = pretty_breaks(n = 7)
  ) +
  labs(
    title = "India's trajectory towards the 2030 targets",
    subtitle = "Historical observations and model-based projections",
    x = NULL,
    y = "Indicator value",
    colour = NULL,
    linetype = NULL,
    caption = paste(
      "Shaded bands: 95% prediction intervals.",
      "Forecast horizons begin in 2024 for MMR and 2025 for the other indicators.",
      "Forecasts extrapolate historical patterns."
    )
  ) +
  theme_sdg() +
  theme(
    legend.position = "bottom",
    panel.spacing = grid::unit(1.2, "lines")
  )
```


India's trajectory towards the 2030 targets
Historical observations and model−based projections
Clean cooking access
Maternal mortality ratio
Safely managed sanitation
Indicator value
100
80
60
40
20
300
200
100
0
75
50
25
2000
2005
2010
2015
2020
2025
2030
2030 target
Forecast
Observed
Shaded bands: 95% prediction intervals. Forecast horizons begin in 2024 for MMR and 2025 for the other indic

5. Interpretation

- **MMR:** the 2030 point forecast is 94.3, against a target of 70. Its prediction interval is 7.0–181.7, indicating substantial uncertainty.
- **Sanitation:** the 2030 point forecast is 76.2%, below the 100% target under the fitted trend.
- **Clean cooking:** the 2030 point forecast is 99.0%, close to but below the 100% target.

The forecast intervals should be interpreted alongside the point forecasts. In particular, an interval extending above 100% for a percentage indicator is a limitation of the unconstrained statistical model, not a physically possible outcome.

6. Limitations

- The analysis extrapolates historical patterns and does not model future policy interventions or structural changes.
- The 2021 MMR observation is retained as verified in the source data. Its influence on forecasts should be acknowledged.
- MMR's last observed year is 2023, whereas sanitation and clean cooking data extend to 2024.
- Forecast intervals depend on model assumptions and may not capture all real-world uncertainty.
- The three indicators have different definitions and units; their numerical values are not directly comparable.

7. Conclusion

The projections indicate that India's three development indicators may follow different trajectories towards 2030. Under the selected models, sanitation remains below its universal-access benchmark, clean cooking approaches its target, and MMR remains uncertain relative to its target.

These results are best interpreted as conditional projections rather than definitive predictions. Continued progress depends on policy implementation, access, data quality and changes that historical time-series models cannot fully anticipate.

Data and references

- **Maternal mortality:** WHO, UNICEF, UNFPA, World Bank Group and UN DESA. *Trends in Maternal Mortality 2000 to 2023* (2025). WHO publication.
- **Safely managed sanitation:** World Bank, World Development Indicators. Indicator SH.STA.SMSS.ZS — India.
- **Clean cooking access:** World Bank, World Development Indicators. Indicator EG.CFT.ACCS.ZS — India.

Forecasts are the author's calculations using the project dataset and R time-series models.